Solution to Assignment 1, 2024

Q1:

(1) Contemporaneous zero restrictions or the LR neutrality restriction.

Please see handwritten notes Lecture 1.

(2) Price(liquidity) puzzle and output puzzle. If shocks to FF are used as the measure of monetary policy actions, no output puzzle was detected, but price puzzle remains. Commodity prices or other asset prices should be included in VAR.

Q2:

The MIU model of section 2.2 implied that the marginal rate of substitution between money and consumption was set equal to $i_t/(1+i_t)$ (see (2.12)). That model assumed that agents entered peirod t with resources ω_t and used those to purchase capital, consumption, nominal bonds, and money. The real value of these money holdings yielded utility in period t. Assume instead that money holdings chosen in period t do not yield utility until period $t+1$. Utility is $\sum \beta^i U(c_{t+i}, M_{t+i}/P_{t+i})$ as before, but the budget constraint takes the form

$$\omega_t = c_t + \frac{M_{t+1}}{P_t} + b_t + k_t, \tag{1.1.1}$$

and the household chooses c_t, k_t, b_t, and M_{t+1} in period t. The household's real wealth ω_t is given by

$$\omega_t = f(k_{t-1}) + (1-\delta)k_{t-1} + (1+r_{t-1})b_{t-1} + m_t. \tag{1.1.2}$$

Derive the first-order condition for the household's choice of M_{t+1} and show that

$$\frac{U_m(c_{t+1}, m_{t+1})}{U_c(c_{t+1}, m_{t+1})} = i_t. \tag{1.1.3}$$

Solution:

The representative household's problem is

$$\max \sum_{t=0}^{\infty} \beta^t U(c_t, m_t) \tag{1.1.4}$$

$$\text{s.t. } c_t + (1+\pi_{t+1})m_{t+1} + b_t + k_t = f(k_{t-1}) + (1-\delta)k_{t-1} + \frac{1+i_{i-1}}{1+\pi_t}b_{t-1} + m_t. \tag{1.1.5}$$

The Bellman equation with Lagrangian multiplier is then given by

$$V(k_{t-1}, b_{t-1}, m_t) = \max_{c_t, m_{t+1}, k_t, b_t} \{U(c_t, m_t) + \beta V(k_t, b_t, m_{t+1})$$
$$+\lambda_t[f(k_{t-1}) + (1-\delta)k_{t-1} + \frac{1+i_{i-1}}{1+\pi_t}b_{t-1} + m_t - c_t - (1+\pi_{t+1})m_{t+1} - b_t - k_t]\} \tag{1.1.6}$$

The first-order conditions with respect to c_t, m_{t+1}, k_t, b_t are given by

$$U_c(c_t, m_t) = \lambda_t, \tag{1.1.7}$$
$$\beta V_m(k_t, b_t, m_{t+1}) = \lambda_t(1+\pi_{t+1}), \tag{1.1.8}$$
$$\beta V_k(k_t, b_t, m_{t+1}) = \lambda_t, \tag{1.1.9}$$
$$\beta V_b(k_t, b_t, m_{t+1}) = \lambda_t. \tag{1.1.10}$$

The Envelop Theorem implies

$$V_m(k_{t-1}, b_{t-1}, m_t) = U_m(c_t, m_t) + \lambda_t, \tag{1.1.11}$$
$$V_k(k_{t-1}, b_{t-1}, m_t) = \lambda_t[f'(k_{t-1}) + (1-\delta)], \tag{1.1.12}$$
$$V_b(k_{t-1}, b_{t-1}, m_t) = \lambda_t \frac{1+i_{t-1}}{1+\pi_t}. \tag{1.1.13}$$

Hence, the first-order conditions can be rewritten as

$$\beta[U_m(c_{t+1}, m_{t+1}) + U_c(c_{t+1}, m_{t+1})] = (1+\pi_{t+1})U_c(c_t, m_t), \tag{1.1.14}$$
$$\beta U_c(c_{t+1}, m_{t+1})[f'(k_t) + (1-\delta)] = U_c(c_t, m_t), \tag{1.1.15}$$
$$\beta U_c(c_{t+1}, m_{t+1})\frac{1+i_t}{1+\pi_{t+1}} = U_c(c_t, m_t). \tag{1.1.16}$$

Using Eq. (1.1.16), we can rewrite Eq. (1.1.14) as

$$\frac{U_m(c_{t+1}, m_{t+1})}{U_c(c_{t+1}, m_{t+1})} = \beta^{-1}(1+\pi_{t+1})\left(\beta\frac{1+i_t}{1+\pi_{t+1}}\right) - 1 = i_t. \tag{1.1.17}$$

Q3:

A commonly used specification of the demand for money, originally due to Cagen (1956), assumes that $m = Ae^{-\alpha i_t}$ where A and α are parameters and i is the nominal interest rate. In the Sydrauski (1976) model with the timing convention outlined in Question 1.1, assume that the utility function is separable in consumption and real balances:

$$u(c_t, m_t) = w(c_t) + v(m_t), \tag{1.2.1}$$

and further assume that

$$v(m_t) = m_t(B - D \ln m_t), \tag{1.2.2}$$

where B and D are positive parameters. Show that the demand for money is given by

$$m_t = Ae^{-\alpha_t i_{t-1}}, \tag{1.2.3}$$

where $A = e^{\frac{B}{D}-1}$ and $\alpha_t = \frac{w'(c_t)}{D}$.

Solution:

By Eq. (1.1.17), the money demand is given by

$$U_m(c_t, m_t) = i_{t-1} U_c(c_t, m_t). \tag{1.2.4}$$

By the given functional form of the utility, we have

$$U_m(c_t, m_t) = v'(m_t) = B - D - D \ln(m_t), \tag{1.2.5}$$

$$U_c(c_t, m_t) = w'(c_t). \tag{1.2.6}$$

Therefore, the money demand function is

$$B - D - D \ln(m_t) = i_{t-1} w'(c_t), \tag{1.2.7}$$

which implies

$$m_t = e^{\frac{B}{D}-1} e^{-\frac{w'(c_t)}{D} i_{t-1}}. \tag{1.2.8}$$

This is indeed equivalent to

$$m_t = Ae^{-\alpha_t i_{t-1}}, \tag{1.2.9}$$

where $A = e^{\frac{B}{D}-1}$ and $\alpha_t = \frac{w'(c_t)}{D}$.

Q4:

Suppose a nominal interest rate of i_m is paid on money balances. These payments are financed by a combination of lump-sum taxes and printing money. Let a be the fraction financed by lump-sum taxes. The government's budget identity is $\tau_t + v_t = i_m m_t$, with $\tau_t = a i_m m_t$ and $v = \theta m_t$. Using Sidrauski's model, do the following:

1. Show that the ratio of the marginal utility of money to the marginal utility of consumption depends on $r + \pi - i_m = i - i_m$. Explain why.

2. Show how $i - i_m$ is affected by the method used to finance the interest payments on money. Explain the economics behind your result.

Solution:

The representative household's problem is

$$\max \sum_{t=0}^{\infty} \beta^t u(c_t, m_t)$$

$$\text{s.t. } c_t + m_t + b_t + k_t = f(k_{t-1}) + (1-\delta)k_{t-1} + \frac{(1+i_{t-1})b_{t-1} + (1+i_m)m_{t-1}}{1+\pi_t} - \tau_t.$$

The Bellman equation is then given by

$$V(k_{t-1}, b_{t-1}, m_{t-1}) = \max_{c_t, m_t, k_t, b_t} \{u(c_t, m_t) + \beta V(k_t, b_t, m_t) + \lambda_t [f(k_{t-1}) + (1-\delta)k_{t-1} + \frac{(1+i_{t-1})b_{t-1} + (1+i_m)m_{t-1}}{1+\pi_t} - \tau_t - c_t - m_t - k_t - b_t]\}. \tag{1.3.1}$$

The first-order conditions with respect to c_t, m_t, k_t, b_t are respectively

$$u_c(c_t, m_t) = \lambda_t, \tag{1.3.2}$$

$$\beta V_m(k_t, b_t, m_t) + u_m(c_t, m_t) = \lambda_t, \tag{1.3.3}$$

$$\beta V_k(k_t, b_t, m_t) = \lambda_t, \tag{1.3.4}$$

$$\beta V_b(k_t, b_t, m_t) = \lambda_t. \tag{1.3.5}$$

The Envelop Theorem implies

$$V_m(k_{t-1}, b_{t-1}, m_{t-1}) = \lambda_t \frac{1+i_m}{1+\pi_t}, \tag{1.3.6}$$

$$V_k(k_{t-1}, b_{t-1}, m_{t-1}) = \lambda_t [f'(k_{t-1}) + (1-\delta)], \tag{1.3.7}$$

$$V_b(k_{t-1}, b_{t-1}, m_{t-1}) = \lambda_t \frac{1+i_{t-1}}{1+\pi_t}. \tag{1.3.8}$$

Hence, the first-order conditions can be rewritten as

$$\beta u_c(c_{t+1}, m_{t+1}) \frac{1+i_m}{1+\pi_{t+1}} + u_m(c_t, m_t) = u_c(c_t, m_t), \tag{1.3.9}$$

$$\beta u_c(c_{t+1}, m_{t+1}) \frac{1+i_t}{1+\pi_{t+1}} = u_c(c_t, m_t), \tag{1.3.10}$$

$$\beta u_c(c_{t+1}, m_{t+1}) [f'(k_t) + (1-\delta)] = u_c(c_t, m_t). \tag{1.3.11}$$

Define $1 + r_t = f'(k_t) + (1-\delta)$, then using Eq. (1.3.10) and (1.3.11), we can derive the Fisher equation

$$1 + i_t = (1+\pi_{t+1})(1+r_t) \text{ or } i_t \approx r_t + \pi_{t+1}. \tag{1.3.12}$$

Then, using Eq. (1.3.11), we can rewrite Eq. (1.3.9) as

$$\frac{u_m(c_t, m_t)}{u_c(c_t, m_t)} = 1 - \frac{1 + i_m}{1 + i_t} \approx i_t - i_m \approx r_t + \pi_{t+1} - i_m. \tag{1.3.13}$$

In the steady-state equilibrium, we have

$$\frac{u_m(c, m)}{u_c(c, m)} \approx i - i_m \approx r + \pi - i_m. \tag{1.3.14}$$

In equilibrium, the marginal rate of substitution between money and consumption is equal to the opportunity cost of money. The nominal interest payment on real money balances reduces the opportunity cost of holding money, and thus is subtracted from the nominal interest on bonds.

The government's budget identity $\tau_t + v_t = i_m m_t$ with $\tau_t = a i_m m_t$ and $v_t = \theta m_t$ implies that $\theta = (1 - a) i_m$. In the steady-state equilibrium, the money growth process $m_{t+1} = \frac{1+\theta}{1+\pi_{t+1}} m_t$ implies $\pi = \theta$ in the steady state. Hence, if a increases, then θ will decrease and so will the steady-state inflation rate. $i - i_m$ will thus decrease due to smaller π. Intuitively, if a larger fraction of interest payment on money is financed by lump-sum taxes, then less money need to be printed, which slows down the money growth and results in a lower inflation and thus lower opportunity cost of holding money.

Q5: (Chapter 2, Question 2.10)

10. Consider the following two alternative specifications for the demand for money given by (2.39) and (2.40).
    a. Using (2.39), calculate the welfare cost as a function of η.
    b. Using (2.40), calculate the welfare cost as a function of ξ.

1. When the money demand takes the form

$$\ln(m) = \ln(A) - \eta \ln(i), \tag{1.4.1}$$

which can be written as

$$m = Ai^{-\eta}, \tag{1.4.2}$$

the welfare cost of inflation, measured by the area under the money demand curve, is calculated by

$$\int_0^i Ax^{-\eta} dx = \frac{Ai^{1-\eta}}{1-\eta}. \tag{1.4.3}$$

2. When the money demand takes the form

$$\ln(m) = \ln(B) - \xi i, \tag{1.4.4}$$

which can be written as

$$m = Be^{-\xi i}, \tag{1.4.5}$$

then the welfare cost of inflation is calculated by

$$\int_0^i Be^{-\xi x} dx = \frac{B}{\xi}(1 - e^{-\xi i}). \tag{1.4.6}$$